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Clean Technology Evaluation Framework

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Takshashila Discussion Document 2026-17
Version 1.0, June 2026

This paper presents a framework to evaluate emerging clean technologies based on their potential to enhance India's energy security and drive economic growth.

Recommended Citation:

Bhumika Sevkani, "Clean Technology Evaluation Framework", Takshashila Discussion Document 2026-17, Version 1.0, June 2026, The Takshashila Institution

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1 Executive Summary

The paper introduces an evaluation framework that assesses emerging clean technologies across seven dimensions: sectoral applicability, substitutability, scalability, export potential, domestic capabilities, resource availability, and supply chain concentration. The primary objective of the framework is to identify emerging clean technologies that India should prioritise to strengthen its energy security and expand economic growth opportunities.

Applying this framework to Solid-State Batteries (SSBs) and Fuel Cell Electric Vehicles (FCEVs) demonstrates that while these technologies show promise in enhancing India's energy security, their adoption is currently hindered by cost, infrastructure, and supply chain challenges. The framework provides a comprehensive profile of a clean technology, highlighting its contribution to India's goals and the capabilities that need to be developed.

2 Introduction

The global energy transition is accelerating the adoption of clean technologies as countries seek to provide affordable, reliable, and sustainable energy while meeting their climate commitments. For India, this transition is closely linked to its decarbonisation objectives and long-term energy security. The country has made significant progress in renewable energy deployment, with non-fossil fuel sources accounting for nearly 53% of its installed electricity capacity as of March 2026. In 2022, India updated its Nationally Determined Contributions (NDCs), targeting 50% of cumulative installed electric power capacity from non-fossil fuel sources by 2030¹—a milestone it had already achieved by June 2025.² However, meeting future goals will require the adoption of a broader range of clean technologies beyond renewable power generation—including energy storage systems, electric vehicles (EVs), green hydrogen, and other emerging technologies that can reduce fossil fuel dependence while supporting economic growth.

At the same time, the transition to clean technologies also introduces new vulnerabilities. Different technologies vary considerably in their ability to substitute fossil fuels, their costs of adoption, and their dependence on imported components and raw materials. India's experience with solar photovoltaics and lithium-ion batteries illustrates these challenges. Despite widespread domestic solar deployment, India remains dependent on imports of photovoltaic (PV) cells and has been unable to fully capitalise on the economic opportunities generated by the global renewable energy boom. Similarly, limited early investments in battery research and development prevented the emergence of a domestic lithium-ion battery ecosystem, resulting in continued

dependence on imported battery cells. The EV sector has followed a similar trajectory, with relatively slow adoption and limited export competitiveness. These experiences highlight the importance of considering not only decarbonisation objectives but also domestic value creation, supply-chain resilience, and geopolitical dependencies when identifying key technologies.

Beyond meeting domestic energy requirements, India must also position itself to capture emerging opportunities in global clean technology markets. The contemporary energy transition has created substantial economic opportunities, prompting countries worldwide to introduce industrial and trade policies in order to strengthen domestic clean technology ecosystems. China's success demonstrates the scale of these opportunities. Through sustained investments across manufacturing, research and development, and supply chains, China has become the dominant global supplier of several clean technologies—including solar PV modules, batteries and electric vehicles. Its first-mover advantage and cost competitiveness have enabled it to capture a significant share of the global demand. In contrast, India's solar PV module exports gained momentum only after the United States increased scrutiny of Chinese and Southeast Asian exports. As demand for technologies such as batteries, new energy vehicles, and green hydrogen continues to expand globally, India must identify areas where it can build competitive advantages and integrate into global value chains, rather than remain primarily an importer of clean technologies.

This challenge underscores the need for a systematic framework to identify which emerging clean technologies should be prioritised for targeted intervention. Given resource constraints and competing policy priorities, it is essential to assess technologies across multiple dimensions—including their contribution to energy security, domestic capabilities across the value chain, geopolitical risks, and global market opportunities. The primary objective of the framework presented in this document is to identify emerging clean technologies that India should prioritise in order to strengthen energy security, and expand economic growth opportunities. While emission reduction remains a fundamental prerequisite for clean technology support, this framework focuses on the economic and strategic dimensions of a technology's adoption and assumes that the technology being assessed will have clear advantages for emissions reduction.

Rather than producing a single definitive ranking, this framework generates a multi-dimensional profile for each technology, thus allowing policymakers and industry to identify trade-offs and complementarities. The framework is applicable to both primary technologies that directly produce renewable energy, and enabling technologies that facilitate renewable energy adoption, such as energy storage systems. Using this approach, the document evaluates the case for Solid-State Batteries (SSBs) and Fuel Cell Electric Vehicles (FCEVs) while providing a tool to support India's long-term energy and industrial strategy.

3 Clean Technology Evaluation Framework

To meaningfully participate in the global value chains for emerging clean technologies, India must identify technologies that not only help it meet its energy security needs but also align with its resource endowment and production capabilities.³ To identify the same, the framework assesses two of the many emerging clean technologies against the below-mentioned criteria. Each criterion has a scoring guide for assessing the value and feasibility of pursuing an emerging clean technology for India. The scores are on a Likert scale, measuring 1-4. Further, the paper also applies the framework on 'Fuel Cell Electric Vehicles' and 'Solid State Batteries' to understand if India should prioritise these technologies. In addition to demonstrating the utility of the framework, the selected technologies are frequently cited for their potential to enhance India's energy security⁴ and independence.⁵ Through this framework, the aim is to assess the contribution of SSBs and FCEVs towards reaching the stated goals.

3.1 Seven Criteria For Scoring:

1. Sectoral Applicability
2. Substitutability
3. Scalability
4. Export Potential
5. Domestic Capabilities
6. Availability of Natural Resources
7. Supply Chain Concentration

Scoring Guides:

1. Sectoral Applicability

This criterion looks at the applicability and utilisation of the technology in various sectors such as transport, power and manufacturing. An assessment based on this criterion should determine the extent to which a clean technology can mitigate oil dependence or conventional fuel consumption in the country. Since the framework is designed for emerging technologies, the applicability is judged based on existing use cases.

Table 1: Scoring for Sectoral Applicability

Description	Score
Widely applicable – in sectors accounting for ≥40% of total oil consumption, and includes at least one dominant sector.	5

Description	Score
Applicable to two or more sectors, and accounting for at least 25% of the total consumption.	4
Applicable to one sector, and accounting for at least 10% of the total consumption.	3
Indirect impact – affects consumption through system improvement instead of fuel substitution.	2
Marginal	1

2. Substitutability

This criterion examines two dimensions of substituting conventional fuels with clean technologies. The framework also aims to capture the applicability of enabling technologies like batteries, which do not directly act as a replacement for conventional fuel in end-use but enable the transition away from conventional fuels. The first dimension in substitution, therefore, is direct or indirect substitution. The second dimension covers the conditions and costs under which the technology substitutes the usage of conventional fuels. This includes the additional distribution and end-use infrastructure required for the technology.

Substitution score:

Direct Substitution: A=2

Indirect Substitution: A=1

Conditions for Substitution:

For direct substitution, if the technology is cost-competitive with conventional fuel at current or near-term projected prices.

OR

If an enabling technology has been proven to enhance the distribution and adoption of a clean technology in the case of enablers: +1

Compatible with existing end-use infrastructure or existing systems in case of enabling technology: +1

Technology has a latent potential to scale without direct government financing support: +1

Table 2: Scoring for Substitutability:

The dominant sector here implies that it is one of the top four oil and gas consuming sectors in India. The top four oil and gas consuming sectors in India include road transport, domestic cooking and heating, petrochemicals and manufacturing.

Description	Score
Direct substitution meets all three conditions.	5
Direct substitution meets two readiness conditions, or the Enabler meets all three readiness conditions.	4
Direct substitution meets one readiness condition, or the Enabler meets two readiness conditions.	3
Direct substitution meets no readiness conditions, or the Enabler meets one readiness condition.	2
The Enabler meets no readiness conditions.	1

3. Scalability

Scalability refers to the deployment timelines for a technology. It evaluates the technology based on its Technology Readiness Level (TRL) and capital intensity, which can significantly affect the time needed for commercial adoption.

Table 3: Scoring for Scalability

Description	Score
Commercial stage (TRL 8-9) and low capital intensity	5
Commercial stage (TRL 8-9) and high capital intensity	4
Pilot stage (TRL 6-7) and low capital intensity	3
Pilot stage (TRL 6-7) and high capital intensity	2
Pre-commercial (TRL 1-5) regardless of capital intensity	1

4. Export Potential:

This criterion looks at the export potential of a technology across its value chain. It borrows from the Global Value Chain literature⁶ and the Smile curve.⁷ Analysing it through the lens of GVCs is helpful in understanding if the technology can be plugged into existing production networks based on the dimensions of complexity and codifiability of transactions, as well as logistical fungibility. The complexity of transaction involves differentiation in products, whereas codifiability involves the standardisation of products and processes. Higher complexity and low codifiability require more exchange of information between firms. Logistical

fungibility, on the other hand, considers difficulties and expenses associated with shipping the product or components. Lower complexity and higher codifiability in transactions enhances the export potential of a product.

Table 4: Scoring for Export Potential

Description	Score
Universal standard, high value-to-weight ratio, can be traded on global exchanges or easily licensed (IP).	5
“Plug-and-play” components that are easy to integrate into foreign systems with minimal custom engineering.	4
Modular tech; specifications are clear, but face logistics or trade barriers.	3
High “tacit” knowledge; export is possible but requires deep, long-term bilateral partnerships.	2
Non-standardised tech, highly customised, and hard to move	1

The smile curve logic asserts that the initial (R&D, and product design) and the final (marketing and after-sales services) stages of any product's value chain have a higher value added than the midstream—manufacturing and assemble stage. Gereffi et. al. explain the different types of value chain governance that emerge from a combination of complexity and codifiability of transactions.

5. Domestic Capabilities

This criterion looks at the existing firm capabilities in India that can impact value chain positioning in the future. Since the framework is for emerging technologies, firm capabilities can be judged based on transferable capabilities from similar technologies.

Table 5: Scoring for Domestic Capabilities

Description	Score
Vertically integrated or Leading in patent publications	5
Manufacturing key components, and Operation & Maintenance capabilities	4
Intermediate (Import key components)	3
Assembly only	2
Importing and Distribution	1

6. Resource Availability

This criterion aims to assess the domestic availability of natural resources required for the production of a clean technology. The availability of the required resources domestically can make the production and scaling of a technology faster and more affordable, given a government's resource constraints. Import dependence, on the other hand, can pose both higher costs and supply chain risks, especially if the import source is concentrated in a single country or geographical location.

Table 6: Scoring for Resource Availability

Description	Score
Proven domestic reserves with active extraction.	5
Domestic resources are confirmed, but extraction requires investment.	4
Domestic reserves present but geographically or seasonally constrained.	3
Substitute resource is domestically available.	2
No domestic resources, import-dependent	1

7. Supply Chain Concentration

The supply chain concentration criterion examines concentration only in China for the purposes of this paper. The framework assesses the extent of Chinese dominance in the supply chain by examining two factors: the concentration of raw material sources, and the proportion of global manufacturing capacity for a key component located in China.

Table 7: Scoring for Supply Chain Concentration

Description	Score
China accounts for <25%	5
China accounts for 25–40%	4
China accounts for 40–60%	3
China accounts for 60–75%	2
China accounts for >75%	1

With China's dominance in clean technology supply chains, India must evaluate its strategy to ensure that its energy dependency does not merely transition from oil imports to essential raw materials or parts managed by a strategic rival.

Lastly, the framework assigns weights to each criterion to help address trade-offs. Technologies do not score uniformly across

criteria. A technology might be highly scalable or have strong export potential, but have low domestic capability. The weights address these contradictions and indicate the significance of different criteria with regard to India's interests. They also ensure that a deficit in one criterion does not disqualify a project if it offers substantial benefits toward meeting India's objectives.

Sectoral applicability and substitutability command the highest weightage, given that they form the basis for primary objectives, and are key to driving other factors like demand in the global and domestic markets. Export potential and domestic capabilities are assigned a weight of 15% each, as they are necessary to meet the secondary objective of leveraging economic opportunities generated by energy transition initiatives.

Table 8: Criteria Weightage

Criteria	Weightage
Sectoral Applicability	20%
Substitutability	20%
Export Potential	15%
Domestic Capabilities	15%
Scalability	10%
Supply Chain Concentration	10%
Resource Availability	10%

4 Applying the Framework to Solid State Batteries

Solid-State Batteries (SSBs) are an emerging alternative to conventional Lithium-ion batteries. SSBs utilise solid electrolytes instead of the flammable liquid or gel electrolytes used in Lithium-ion batteries—offering advantages in thermal stability and safety, energy density and charging cycles.⁸ Increased adoption of EVs and renewable energy technologies globally is also driving up the demand for safe and high-performance energy storage solutions. SSBs are expected to fulfil this demand for long-range EVs that can provide up to 600-800km on a single charge, and for energy storage systems required to improve the reliability of renewable energy projects, which are otherwise intermittent. The variable nature of renewable energy power generation limits its adoption; advances in energy storage systems are crucial in addressing this limitation and transitioning away from fossil fuels. SSBs allow for greater energy storage, while utilising the same space due to their higher energy density (300–500 Wh/kg) as compared to Li-ion batteries (150–250

Wh/kg). So, SSBs can contribute to grid stability by storing the surplus energy produced during peak hours, and releasing it during periods of energy deficit.⁹

The growing interest in this battery technology is also evident in the increasing number of patents over the past decade. Patent families related to SSBs have increased significantly, rising from 290 in 2010 to 2,033 in 2023. Globally, Japan is the leader in this technology, accounting for nearly 40% of all SSB patent families published between 2000 and 2023. Inventors from Japan are responsible for publishing more than 7,000 patent families during this period.¹⁰ However, several Chinese firms are also closely following this lead.

While global interest in the technology is evident in patents and investments by leading battery manufacturers, it has yet to achieve commercial viability. But it could offer significant opportunities for emerging economies once the technology scales. Wider applications in EV, medical devices, consumer electronics, and grid-level energy storage offer a large market. While the applications look promising, any emerging technology comes with the challenge of high initial costs as the manufacturing processes and supply chains are yet to reach efficiency. SSBs, therefore, can be limited to premium markets at first. Once margins for manufacturers reach a point where they can scale production and reduce costs—which is unlikely to be the case until the first half of the 2030s—markets can open up a greater number of actors.¹¹ Nonetheless, this would be the right time for emerging economies to lock in the leverage that the technology can provide once it scales.

A patent family is a collection of patent applications covering the same or similar technical content, thereby protecting the same invention across different countries.

Table 9: Evaluating SSBs

Criteria	Description	Score
Sectoral Applicability	Indirect impact (affects consumption through system improvement instead of fuel substitution).	2
Substitutability	The Enabler meets all three readiness conditions.	4
Scalability	TRL 6, High Capital Intensity	2

Criteria	Description	Score
Export Potential	High “tacit” knowledge; export is possible but requires deep, long-term bilateral partnerships.	2
Domestic Capabilities	Assembly only (Borrowing from Li-ion)	2
Resource Availability	Domestic resources are confirmed, but extraction requires investment.	4
Supply Chain Concentration	China accounts for 60–75%	2

In terms of substitutability, pilots and prototypes for SSBs have shown that they can make clean technologies like Electric Vehicles more efficient by enabling longer range and increased lifecycles. They could also be helpful in reducing buyer anxiety related to frequent charging requirements of EVs, thereby increasing adoption rates. To this end, two companies, Chery and Sunwoda, have recently unveiled advancements in solid-state battery technology, signalling significant private sector interest and investment in the field.¹² Chery introduced a solid-state battery module boasting an impressive energy density of 600 Wh/kg, which is projected to enable an EV range of 1,300 km—twice the range provided by conventional lithium-ion batteries. Similarly, Sunwoda presented a polymer solid-state battery. This battery features an energy density of 400 Wh/kg and offers 1,200 cycles. This performance could translate into EV ranges exceeding 1,000 km. Apart from changes required in battery and thermal management softwares, the technology is largely compatible with the existing end-use infrastructure of EVs and RE projects. A higher value for the private sector in improving end-use systems, and reducing the costs of other components in these systems, provides it with the potential to scale without direct government support in the future. The possibility of SSBs becoming modular with improvements in manufacturing capabilities adds to their potential to scale.

On Scalability, the International Energy Agency (IEA) rates SSBs at ‘Technology Readiness’ Levels of 6 with demonstrated applications in EVs and humanoid robots.¹³ However, the technology requires high capital intensity for manufacturing and has higher final costs than Lithium-ion batteries.** **Common sulfide electrolytes used in SSBs are moisture-sensitive, thereby requiring strict control and specialised dry room facilities.¹⁴

Since the manufacturing processes are complex and require substantial tacit knowledge in building manufacturing lines, the export potential is currently limited.¹⁵ It will require a long-term partnership to establish a presence in a manufacturer's supply chain, since it is not simply a 'plug-in' component and requires certain modifications for smooth integration with a device—especially in battery and thermal management softwares. Further, although the supply chain is nascent, it is likely to be concentrated in China, as the key materials required for a solid-state battery include refined lithium, nickel and cobalt.¹⁶ Except for Nickel, production capacity for refined lithium and nickel is concentrated in China, with 73% of the global production of refined lithium and 80% of the global production of Cobalt occurring in the country. Capacity for nickel, on the other hand, is split between China and Indonesia.¹⁷ In terms of domestic reserves, India has some Lithium reserves, but extraction will take time. While the Jammu reserves are at the G3 stage and lack private-sector interest in extraction¹⁸ the Chhattisgarh mines are close to operationalisation.¹⁹ For cobalt and nickel, India has discovered resources of about 45 million tonnes and 4.7 million tonnes respectively, but they are yet to be explored.²⁰

Moreover, the manufacturing capacity of SSBs itself is also likely to be concentrated as 80% of the global manufacturing capacity for solid-state batteries—that is currently installed, under construction, or has reached a final investment decision—is located in China.

Lastly, the development of SSBs faces multiple structural constraints across materials, processes, standards, and costs.²¹ On the materials side, limited lithium sulphide production capacity and its high cost remain key challenges. Additionally, carbon anodes cost nearly four times as much as artificial graphite. With regards to processes, dry manufacturing technologies remain immature, limiting the capabilities of firms to scale production rapidly. SSBs also lack global testifying standards, which induces further uncertainty for adoption. All of these challenges add up to the cost, with all-solid-state batteries estimated to be six to eight times more expensive than traditional liquid lithium-ion batteries as of 2025.²²

The final score for SSBs, considering the criteria weightages, is 52 out of 100—entirely pulled up by substitutability and domestic reserves that are yet to be commercialised. This closely reflects SSBs' potential as a strong enabler and further supports the conclusion that India currently lacks the industrial foundations and export capabilities to capitalise on it, while supply chain risks remain significant.

Table 10: Evaluating FCEVs

Stages of Mining Exploration include G4 (Reconnaissance): Broad search to identify areas with high mineral potential, G3: Confirmed deposits, G2: Determining the shape, size and grade of deposits, and G1 comprises detailed exploration after which the deposits can begin commercial extraction.

Criteria	Description	Score
Sectoral Applicability		5
Substitutability	Direct substitution meets no readiness conditions.	2
Scalability	TRL 9, High Capital Intensity	4
Export Potential	High “tacit” knowledge; export is possible but requires deep, long-term bilateral partnerships.	2
Domestic Capabilities	Intermediate (Borrowing from PEM Electrolyser manufacturing)	3
Resource Availability	No domestic reserves, import-dependent	1
Supply Chain Concentration	China accounts for 40–60%.	3

Hydrogen Fuel Cell Electric Vehicles (FCEVs) are regarded as the next game-changer for automotive technologies after Battery Electric Vehicles (BEVs). Powered by hydrogen, FCEVs utilise a propulsion system that converts stored hydrogen energy into electricity using a fuel cell, similar to how electric vehicles operate. It also uses a small battery pack that uses electricity to power the motor. This process makes FCEVs more efficient than conventional Internal Combustion Engine (ICE) vehicles, with no harmful emissions. Key components include the fuel cell stack, with the most common one being the Proton Membrane Exchange (PEM) fuel cell, and the Lithium-ion battery pack.

Hydrogen’s applications are most focused on automotive solutions globally. Countries are either running pilots or have already deployed FCEVs in passenger cars, light commercial vehicles, heavy-duty trucks and urban transit buses. Its applications have a greater advantage in large vehicles owing to faster refuelling times – similar to that of diesel engines. BEVs are more competitive in the passenger car segment than FCEVs due to existing infrastructure and lower costs. However, the passenger cars segment is expected to grow in 2026, accounting for 53.51% of the total vehicle market.²³

Since hydrogen can be produced locally, FCEVs make a strong case for meeting India’s energy security goals. A pilot by the

National Institute for Solar Energy is underway to assess the feasibility of Toyota's Mirai on Indian roads and to understand the need for supporting infrastructure, such as refuelling stations and transporting hydrogen.

With regards to sectoral applicability, road transport is a significant consumer of oil and gas in India, and supports both personal mobility and widespread economic activity. If the technology is successful in light and heavy-duty commercial vehicles, it can also contribute to a significant reduction in vehicular emissions from diesel trucks, and result in lower costs of transportation in the long term. Despite having a higher total cost of ownership compared to BEVs or diesel trucks, heavy trucks remain a promising segment in road transport for FCEV applications, with companies like TATA already rolling out various models.²⁴ From the petroleum products consumption, high-speed diesel oil accounts for 38%, whereas petrol accounts for 16%, with most of these products²⁵ being used up in road transport.²⁶

In terms of substitutability, FCEVs can directly replace conventional fuel consumption by ICEs but meet no readiness conditions of being cost-competitive, compatible with existing infrastructure, and requiring no government policy support to scale. At current levels of deployment, FCEVs have a higher total cost of ownership than BEVs and ICEs.²⁷ Driving a petrol car is significantly more expensive, costing approximately INR 8–10 per kilometre. In comparison, hydrogen-fueled vehicles cost about INR 4 per kilometre, while electric vehicles offer the lowest running cost at around INR 0.80 per kilometre.²⁸ BEVs have a higher initial purchase price than the other two types because of batteries, but lower total costs of ownership due to low electricity costs. FCEV passenger cars in India will only be available in the premium segment for the initial years. Toyota Mirai, for instance, which is expected to be launched soon, will cost around INR 60 Lakhs.

For commercial heavy-duty, long-haul transport, FCEVs are projected to become more cost-effective than diesel by 2040. They show strong potential to replace diesel but are not expected to achieve cost parity with Battery Electric Trucks (BETs), even for long-haul routes, unless there is a reduction in Hydrogen costs. However, the projected reduction in Hydrogen costs for FCEVs is expected to become cost-competitive with BEVs between 2030 and 2040, with the total cost of ownership likely being half that of diesel vehicles, thereby indicating significant economic benefits.²⁹ Currently, the production costs of Green Hydrogen is at about \$4-\$5/Kg, and the target under India's National Green Hydrogen Mission is to bring it down to \$2/kg, which will affect the competitiveness of FCEVs significantly.³⁰

In addition to that, the technology requires new infrastructure in terms of refuelling stations, hydrogen storage and transportation facilities, which is highly capital-intensive. The technology is also currently heavily dependent on subsidies across countries

for both FCEVs and hydrogen fuel. Governments are also subsidising the costs for hydrogen production infrastructure until it reaches the stage of market viability. While these might be phased out in the future, governments prioritising the technology indicate solid demand for downstream applications of hydrogen—especially FCEVs. The infrastructure requirements also affect the technology's Scalability score, despite FCEVs reaching TRL 9.³¹ Widespread adoption is also limited by the lack of universal standardisation of the technology and the different approval processes followed in different countries.³² This also affects its export potential, including components such as hydrogen fuel storage tanks in the vehicles.

For Scalability, TRL levels are at 9 with high capital intensity. However, that is only the case due to the new infrastructure required for refuelling stations, hydrogen storage, and transportation tanks.³³ Currently, fuel cell vehicles also cost more, but have the potential to reach cost parity as supply chains develop and localise. Standardisation challenges and regulatory approval processes further delay infrastructure rollout. These limitations restrict widespread adoption despite technological readiness.³⁴

In 2024, there were more than 70,000 fuel cell passenger cars on the roads, globally. In terms of available models, approximately 50 fuel cell commercial vehicle models were available globally, which is 10 times fewer than battery-electric models. Over 60% of these were heavy-duty trucks from roughly 20 manufacturers, while some were facing insolvency due to low market demand. However, the global stock of fuel cell buses grew by about 25% in 2024, with nearly 75% of it being in China, where urban transit is being increasingly revamped, mostly reliant on government procurement. With about 1,700 fuel cell busses, accounting for over 15% of the global stock, Korea also witnessed an uptake in the technology's adoption.³⁵

In terms of export potential, the technology includes components that can be exported globally once it scales up and standards are adopted worldwide. Currently, however, the manufacturing processes involve high tacit knowledge, including that for the development of PEM stacks. Partnerships are essential to develop vehicles compatible with a country's hydrogen ecosystem, while costs remain high due to a lack of localised supply chains. Therefore, high system costs and technical complexity remain key challenges in the market. Furthermore, fuel cell stacks, hydrogen storage tanks, and power electronics require advanced materials, precision manufacturing, a skilled workforce and maintenance expertise—making manufacturing in India reliant on foreign players.

Domestic firm capabilities for manufacturing in India are currently limited to Balance of Plant (BoP) components that are auxiliary to PEM fuel cells. India already manufactures all the BoP components—such as power converters and heat

exchangers needed for electrolyzers—as these are used in other existing applications and can be easily adapted. However, key components for PEMs—such as the Nafion membranes and carbon papers—are not yet developed, and the country is import-dependent for the same. There is a strong potential for localising the manufacturing ecosystem for key components as well.³⁶ The import dependence might then only remain restricted to key raw materials such as platinum and iridium that are not available in India, unless alternatives emerge in the long term. Another key component is the Li-ion batteries, for which India remains import-dependent on China for Lithium-ion cells, as Indian manufacturers possess only assembly capabilities.

The heavy use of Platinum Group Metals (PGM) and Titanium in PEM fuel cells also restricts the ability of the technology to score higher in resource availability. While titanium is available and refined in India, it is currently insufficient, with most of it being used for defence and aerospace applications.³⁷ In terms of supply chain concentration of materials and components, the raw materials are produced across South Africa, Zimbabwe, Russia, Canada and the United States—leaving out concerns of China's dominance in this segment.³⁸ The availability of Iridium is a persistent issue globally, and companies are focusing on reducing the quantities of the metal used in PEMs. For key components, as PEM fuel cells are an emerging technology, they are not yet concentrated in China, with Japan and South Korea leading in IP and manufacturing.³⁹ This offers India a window to rapidly seek partnerships and secure a position in the supply chain before China's manufacturing advantage starts kicking in. Japan leads in PEM fuel cell technology, with Toyota and Honda pioneering automotive uses, supported by government subsidies and research. China currently accounts for approximately 42% of PEM membrane production capacity globally, and is rapidly developing vertically integrated supply chains through tax incentives and aggressive government goals for fuel cell vehicles.⁴⁰

The final score of FCEVs using the evaluation framework is 59 out of 100. Sectoral applicability (20) and domestic capabilities (9), although limited, contribute highly to the score, whereas substitutability (8.0) significantly drags it down. This reinforces the fact that though FCEVs are commercially ready, they are currently uncompetitive without infrastructure and cost conditions being met. Resource Availability at 2.0 is the weakest contributor, which can shift import dependence from oil to raw materials. However, since the technology is at TRL 9 and as the infrastructure develops further, FCEVs are sure to enhance their export potential. Meanwhile, projections state that India will also develop further domestic capabilities in component manufacturing.

Since the framework evaluates a technology's potential rather than its proven performance in emission reduction and technical feasibility, any technology that scores above the range of 35-40 can be considered to have significant potential for India's energy

transition. While the score can provide an initial screening for technologies, it is important to look at the complete profile of a particular technology in order to understand its contribution to India's goals of energy security and economic growth, as represented by the criteria weightages. Besides that, the profile can also point to the gaps that India needs to address vis-à-vis that technology to improve its feasibility.

5 Conclusion

When applied to solid-state batteries and fuel cell electric vehicles, the Clean Technology Evaluation Framework helps analyse an emerging technology's supply chains. Thereby, it contributes to understanding the technology beyond the hype—considering market feasibility, potential dependencies and the opportunities that it can develop for India. It also highlights specific areas, such as enhancing domestic component capabilities or engaging in global standardisation. Ultimately, these efforts could help India leverage emerging clean energy technologies to reduce its heavy dependence on fossil fuel imports while strengthening energy security.

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